PYTHON ACTIVITY-BASED LEARNING ASSIGNMENT

1. COVER PAGE

Title: Python Activity-Based Learning Assignment

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Branch and Section: cse-data science(csd-3)

Subject: Python Programming

College Name: NRI University

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2. TABLE OF CONTENTS

1. Progressive Electricity Bill
2. ATM Transaction Sequence Simulation
3. Student Performance Trend Analysis
4. Multi-Product Shopping Budget
5. Repeated Transaction Detection
6. Password Policy Validation
7. Examination Answer Evaluation
8. Text Compression Using Character Runs
9. Student Marks Matrix Analysis
10. Food Waste Monitoring System

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3. CASE STUDIES

Each case study includes the following sections:

1. Problem Statement
2. Objectives
3. Assumptions
4. Algorithm

5. Flowchart
6. Pseudocode
7. Python Source Code
8. Test Cases and Expected Outputs
9. Output Screenshots
10. Conclusion

All 10 case studies are implemented using Python programming.

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4. FINAL CONCLUSION

This assignment helped me understand Python programming concepts such as variables, conditional statements, loops, lists, strings, and input validation. By completing these case studies, I gained practical experience in solving real-world problems using Python. This activity improved my logical thinking and programming skills.